



Duck Bank

Short loans to named shops on the XRP Ledger.
One shop per vault. Lenders hold the vault MPT share.



QAK

Duck Bank on the XRP Ledger

QAK: 1,000,000,000 supply, issued on FirstLedger in Q4 2026. Issuer blackholed. Issuer half in two escrows.

Lending: built on XLS-65 (Single Asset Vault) and XLS-66 (Lending Protocol). Mainnet deposits open only after SingleAssetVault, LendingProtocol and LendingProtocolV1_1 are enabled on mainnet. On 8 October 2026 none of them is.

<https://trustline-tan.vercel.app> · Not audited. Not an offer.

1. The desk

Duck Bank lends RLUSD to a named shop for a short, fixed term: stock for a season, a counter, a delivery. Each shop gets its own closed-ended vault and its own loan broker on the XRP Ledger. Lenders deposit RLUSD into that vault during its subscription window and receive **the vault MPT share**, the receipt for that shop only. If the shop misses, the loss is written down on that vault. The next shop is a different vault.

QAK is the app token. It lists a shop, sets the fee cut, orders the deposit queue and carries one vote per token on the listing and the cap. These are rules of the Duck Bank app. The ledger knows nothing about them, and no QAK ever stands as loan cover.

2. The ledger model

Every field, flag and error below is taken from the XLS-65 and XLS-66 specifications (including the closed-ended vault extension and LendingProtocolV1_1).

#	Transaction	What Duck Bank sets
1	<code>VaultCreate</code>	Asset RLUSD; <code>VaultKind</code> 1 (ClosedEnded) with <code>SubscriptionDate</code> and <code>RedemptionDate</code> ; <code>AssetsMaximum</code> = the cap; <code>Scale</code> 6 (IOU range 0-18); flag <code>tfVaultShareNonTransferable</code> . Shop name, city and purpose stay off-ledger.
2	<code>LoanBrokerSet</code>	On that <code>VaultID</code> : <code>DebtMaximum</code> = the cap, <code>CoverRateMinimum</code> 15000 (15%), <code>CoverRateLiquidation</code> 100000 (100%), <code>ManagementFeeRate</code> 1000 or 750 (fixed once the broker exists).
3	<code>LoanBrokerCoverDeposit</code>	RLUSD first-loss cover. Only the broker owner can deposit, and only the vault asset (<code>tecWRONG_ASSET</code> otherwise).
4	<code>VaultDeposit</code>	Only while <code>now</code> ≤ <code>SubscriptionDate</code> ; later deposits fail <code>tecEXPIRED</code> . Withdrawals fail <code>tecT00_S00N</code> until <code>RedemptionDate</code> .
5	<code>LoanSet</code>	Only in the investment window. Signed by the broker and by the shop (<code>CounterpartySignature</code> ; without it <code>temBAD_SIGNER</code>). <code>StartDate</code> + <code>PaymentInterval</code> × <code>PaymentTotal</code> + 60 s must be ≤ <code>RedemptionDate</code> . <code>GracePeriod</code> ≤ <code>PaymentInterval</code> .
6	<code>LoanPay</code>	Fixed installments. A payment below the periodic payment fails <code>tecINSUFFICIENT_PAYMENT</code> ; a late one needs <code>tfLoanLatePayment</code> .
7	<code>LoanManage</code>	<code>tfLoanImpair</code> once a payment is overdue; <code>tfLoanDefault</code> once <code>now</code> > <code>NextPaymentDueDate</code> + <code>GracePeriod</code> .
8	<code>LoanDelete</code>	When <code>PaymentRemaining</code> is 0: the loan is settled or written off.

One vault serves one shop, and never a second. The vault's pseudo-account holds the RLUSD and issues the shares. It cannot receive ordinary payments, so money reaches lenders only through `VaultDeposit`, `LoanPay` and, on default, cover moved by `LoanManage`.

The lender holds the vault MPT share. It is non-transferable, redeemable with `VaultWithdraw` from `RedemptionDate`, and it tracks that one vault's assets, including any loss written down on it.

3. The first pool

One shop, a 10,000 RLUSD cap, short term: Kostas Bakery, Athens, stock. Terms: `InterestRate` 10000 (10% a year), `PaymentTotal` 3, `PaymentInterval` 30 days, `GracePeriod` 7 days. The vault and broker have not been created yet, so there is no `VaultID` or `LoanBrokerID` to publish.

Maximum principal under the cap

`AssetsMaximum` = `DebtMaximum` = 10,000 RLUSD. XLS-66 counts the expected interest toward both limits. `LoanSet` fails if `AssetsTotal` $\geq$ `AssetsMaximum`, if `AssetsTotal` + `InterestDue` > `AssetsMaximum`, or if `DebtTotal` + `Principal` + `InterestDue` > `DebtMaximum` (all `tecLIMIT_EXCEEDED`). A full 10,000 deposit plus a 10,000 principal therefore cannot be lent.

Item	Value	How
Periodic payment	$3,333.73 \times 3$	Amortized payment, periodic rate = $10\% \times 30 \text{ days} / 365 \text{ days}$ (XLS-66 A-2)
Maximum principal	9,839.02 RLUSD	Largest principal with <code>principal</code> + <code>InterestDue</code> $\leq$ 10,000, rounded down to the cent with a 0.01 margin
InterestDue to the vault	160.96 RLUSD	Gross interest less the management fee, at fee rate 750 (the lower fee gives the larger <code>InterestDue</code> , so the bound holds for both tiers)
Principal + InterestDue	9,999.98 RLUSD	$\leq$ <code>AssetsMaximum</code> and $\leq$ <code>DebtMaximum</code>
Deposit target	9,839.02 RLUSD	The app stops inviting deposits at the principal
Cover before <code>LoanSet</code>	1,500.00 RLUSD	$(\text{DebtTotal} + \text{Principal} + \text{InterestDue}) \times \text{CoverRateMinimum}$ 15%

The vault is public. The ledger itself accepts deposits up to `AssetsMaximum`, so direct deposits can push `AssetsTotal` to 10,000. In that case `LoanSet` fails and lenders wait for `RedemptionDate` to withdraw.

4. When a shop misses

There is no on-chain collateral and no liquidation. The XLS-66 security section states that the protocol does not offer on-chain algorithmic protection against default. What the ledger does, after the grace period, is `LoanManage` with `tfLoanDefault`:

`DefaultCovered` = $\min(\text{DebtTotal} \times \text{CoverRateMinimum} \times \text{CoverRateLiquidation}, \text{DefaultAmount}, \text{CoverAvailable})$
Vault loss = `DefaultAmount` – `DefaultCovered`, written down on that vault only.

Worked on the first pool, if nothing is repaid: `DefaultCovered` = $\min(9,999.98 \times 15\% \times 100\%, 9,999.98, 1,500.00) = 1,500.00$ RLUSD of cover moved into the vault, and 8,499.98 RLUSD written down on that vault. No other vault is touched. Interest is booked into `AssetsTotal` at `LoanSet`, before it is paid. Redemption still opens on `RedemptionDate` even if the loan is late.

5. QAK

Role	App rule
Listing	A shop locks 250,000 QAK to be listed. This is an app lock.
Fee cut	The app reads the shop's QAK balance on the ledger when the broker is created: at least 100,000 QAK sets <code>ManagementFeeRate</code> 750 (0.75% of interest), otherwise 1000 (1.00%). The rate is fixed after creation.
Queue	Locked QAK orders the Duck Bank deposit queue. It does not reserve a fill; the vault is public during Subscription.
Vote	One QAK, one vote, on the listing and the cap. A vote cannot move another vault's deposits.

Any QAK escrow or lock is an app lock. It is not `CoverAvailable`: cover accepts only the vault asset, RLUSD. On default the ledger takes RLUSD cover only.

6. Tokenomics

QAK was created on FirstLedger in Q4 2026. Supply is 1,000,000,000, fixed.

Bucket	Of supply	Tokens	Use
Public pool	50%	500,000,000	Sold into the opening FirstLedger pool.
Team	10%	100,000,000	McQAK, Donald QAK, Della and Huey QAK. From the escrowed issuer half.
Marketing	20%	200,000,000	Listings, content and the shop desk. From the escrowed issuer half.
AMM	20%	200,000,000	Added to the QAK/XRP AMM after unlock. From the escrowed issuer half.

Launch

FirstLedger, 50 XRP starting liquidity, 1% AMM fee, 15-minute anti-sniper.

Issuer and escrows (read live from the ledger on 8 October 2026)

The issuer `r98RkKUasH5vA3mshD5Bi3eAVHCYwziY9M` is blackholed: master key disabled, regular key set to the blackhole address `rrrrrrrrrrrrrrrrrrrrBZbvji`, no signer list. No more QAK can be issued. The issuer half, 500,000,000 QAK, is in two escrows:

Escrow	Tokens	Unlocks (FinishAfter)
1	125,000,000 QAK	8 October 2026, 15:49 EEST
2	375,000,000 QAK	7 November 2026, 16:24 EET

7. Status and gating

XLS-65/66 need three amendments: `SingleAssetVault`, `LendingProtocol` and `LendingProtocolV1_1`. A rippled release is not activation: an amendment counts only once validators enable it on that network. The app's Status page runs the read-only `feature` request on each network. On 8 October 2026: mainnet none enabled, testnet none enabled, devnet all three enabled.

- **Mainnet:** vault lending is off in the build itself, regardless of amendment status. No mainnet deposits.
- **Testnet:** status only.

- **Devnet:** the real transactions, once all three amendments read enabled. RLUSD does not exist on devnet, so the app uses a labelled test IOU from a configured test issuer; with none configured, devnet actions stay disabled.

Xaman is the wallet. Its network list includes DEVNET, so sign requests can be forced to devnet. Xaman support for the XLS-65/66 transaction types is not verified, and Xaman does not document producing a `LoanSet CounterpartySignature`, so the shop's counter-signature cannot yet be collected in the app. Without an API key, the app runs a wallet labelled mock that signs nothing.

8. Risks

- **Clawback.** The RLUSD issuer has clawback enabled. It can claw back RLUSD from a vault and broker cover above the minimum.
- **No on-chain collateral.** Loans are unsecured on the ledger.
- **No liquidation.** On default only RLUSD cover, up to the rates, reaches the vault. The rest is written down on that vault.
- **Lock-up.** Withdrawals are blocked from `SubscriptionDate` to `RedemptionDate`.
- **Interest booked at LoanSet.** The vault shows expected interest before it is paid.
- **Software.** The code is not audited.

9. Team

The 10% team bucket covers four published roles.

McQAK, Founder

The desk: which shop is listed, the cap and the terms.

Donald QAK, CTO

The app, the Xaman connection and the vault path.

Della, Marketing

The channel, the shop listings and the public note when a pool opens or repays.

Huey QAK, Developer

The pool page, the listing lock check and the test signatures.

10. Roadmap

When	Step	What
Q4 2026	Mint (done)	QAK issued on FirstLedger: 1,000,000,000 supply, 50 XRP starting liquidity, 1% AMM fee, 15-minute anti-sniper. Issuer blackholed; issuer half in two escrows.
Next	Devnet vault + broker	One shop: a closed-ended vault and a loan broker on devnet with a labelled test IOU, cap 10,000.
After the three amendments are enabled on mainnet	Mainnet deposits	Mainnet deposits open only once SingleAssetVault, LendingProtocol and LendingProtocolV1_1 are all enabled on mainnet.

11. References

- XLS-65 Single Asset Vault, including the closed-ended vault extension (XLS-65.1.4).
- XLS-66 Lending Protocol, including LendingProtocolV1_1.
- App: <https://trustline-tan.vercel.app> (Status page: live amendment check). Venue: FirstLedger. Wallet: Xaman.

This paper describes software and token rules. It is not an offer of securities or a promise of returns. Ledger facts were read on 8 October 2026.